WP Lab 7
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Mini Project Synopsis/Proposal

Team members
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Title Financial Instrument Analyser

The objective of this project is to design and develop a web based financial instrument analyser that fetches stock market data using data folder saved locally on machine, performs candlestick and volume based analysis and provided actionable **Buy/ Sell / Hold** signals.

System & Design

User interface (Streamlit Web app)
|
User Stock Search
|
Fetch Market Data (Locally or through API)
|
Data Preprocessing & Cleaning (if required)
|
Technincal Analysis Engine
|
Signal Generation (Buy/Sell/Hold)
|
Visualization (Candles + Volume)

Components

1. **Frontend Layer (Streamlit UI)**
    - Stock search bar
    - Timeframe selection (1D, 1W, 1M, 1Y)
    - Interactive charts
    - Signal output panel
2. **Data Fetching Module**
    - Fetches real-time or near real-time OHLCV data
    - Uses financial APIs such as **Yahoo Finance**
3. **Analysis Engine**
    - Candlestick pattern recognition
    - Volume trend analysis
    - Technical indicators computation
4. **Decision Engine**
    - Rule-based logic for Buy / Sell / Hold recommendations

Key Features

1. Stock Search & Real-Time Data Fetching

- Users can search stocks by **ticker symbol or company name**
- Automatically fetches **Open, High, Low, Close, Volume (OHLCV)** data

2. Candlestick Chart Visualization

- Interactive candlestick charts
- Clearly visualizes bullish and bearish price movements
- Supports multiple timeframes

3. Volume Analysis

- Displays volume bars aligned with price action
- Detects:
    - Volume spikes
    - Volume divergence

        - High-volume breakouts

 4. Technical Indicators

- Moving Averages (SMA / EMA)
- RSI (Relative Strength Index)
- MACD
    - Price trend
    - Volume confirmation
    - Indicator thresholds
- Generates a **human-readable trading suggestion**

Example:

- *"High volume breakout above resistance with bullish RSI – BUY signal"*
        - **Simple & Interactive Web Interface**
- Built using **Streamlit**
- No login required
- Clean UI for non-technical usersSupport and Resistance levels

 5. Buy / Sell / Hold Recommendation

- Combines:
    - Price trend
    - Volume confirmation
    - Indicator thresholds
- Generates a **human-readable trading suggestion**

Example:

"High volume breakout above resistance with bullish RSI – BUY signal"

Uniqueness & Innovation

This project is **not just a chart viewer**. Its uniqueness lies in:

✅1. Volume + Candlestick Combined Logic

Most beginner tools show either:

- Price charts OR
- Indicators

This system **correlates price movement with volume behavior**, which is closer to **professional technical analysis**.

✅2. Automated Decision Explanation

Instead of simply showing indicators, the system:

- Explains **why** a Buy or Sell signal is generated
- Converts raw numerical indicators into **plain English insights**

✅3. Educational Focus

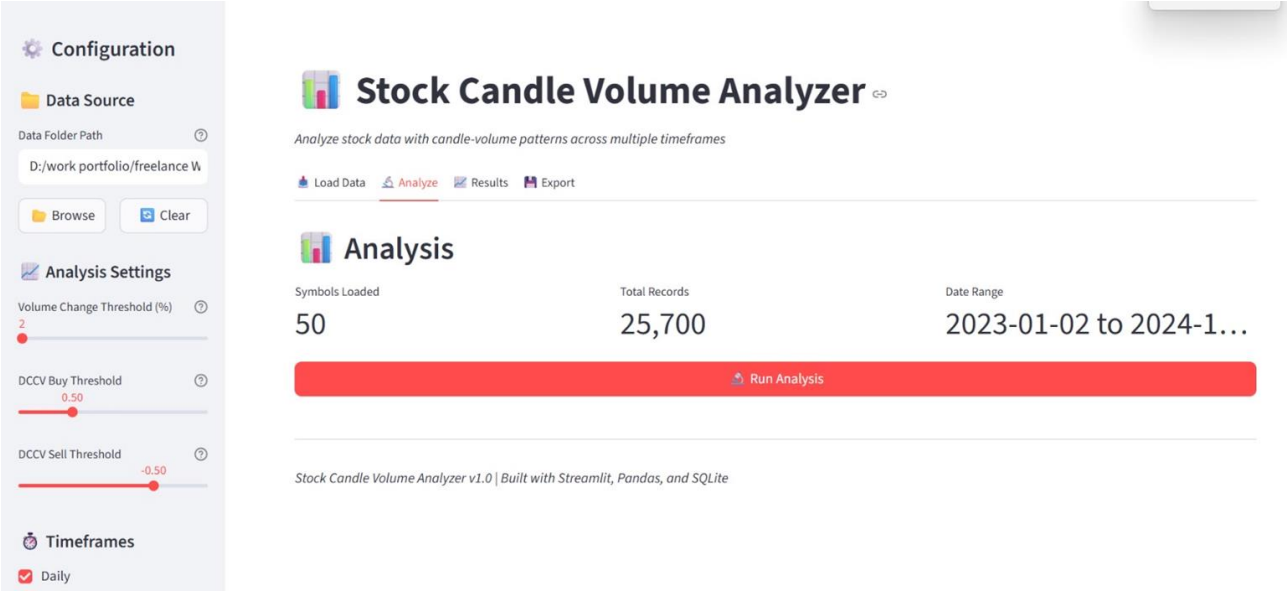
Designed for:

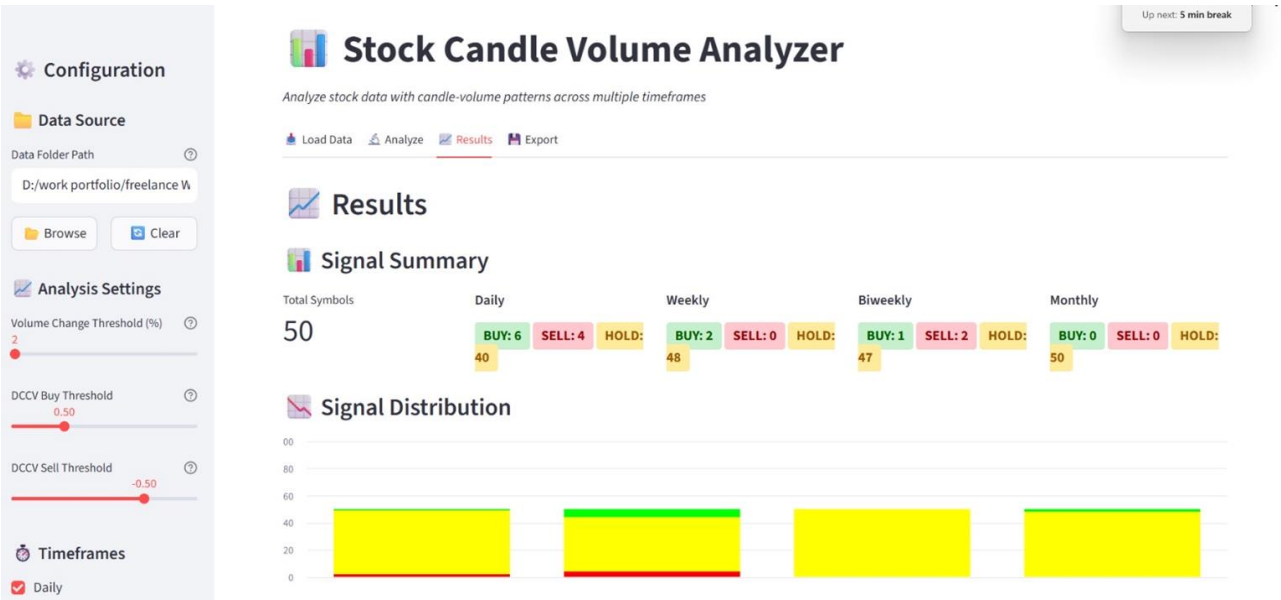
- Beginners learning technical analysis
- Students understanding market behavior
- Not aimed at high-frequency trading bots

✅4. Lightweight & Open Architecture

- Runs as a single web application
- No heavy backend or databases required
- Easily extendable for new indicators

prototype screenshots


Configuration
Data Source
Data Folder Path
D:/work portfolio/freelance W
Browse
Clear
Analysis Settings
Volume Change Threshold (%)
2
DCCV Buy Threshold
0.50
DCCV Sell Threshold
-0.50
Timeframes
Daily
Stock Candle Volume Analyzer
Analyze stock data with candle-volume patterns across multiple timeframes
Load Data
Analyze
Results
Export
Analysis
Symbols Loaded
50
Total Records
25,700
Date Range
2023-01-02 to 2024-1…
Run Analysis
Stock Candle Volume Analyzer v1.0 | Built with Streamlit, Pandas, and SQLite


Up next: 5 min break
Configuration
Data Source
Data Folder Path
D:/work portfolio/freelance W
Browse
Clear
Analysis Settings
Volume Change Threshold (%)
2
DCCV Buy Threshold
0.50
DCCV Sell Threshold
-0.50
Timeframes
Daily
Stock Candle Volume Analyzer
Analyze stock data with candle-volume patterns across multiple timeframes
Load Data
Analyze
Results
Export
Results
Signal Summary
Total Symbols
50
Daily
BUY: 6
SELL: 4
HOLD: 40
Weekly
BUY: 2
SELL: 0
HOLD: 48
Biweekly
BUY: 1
SELL: 2
HOLD: 47
Monthly
BUY: 0
SELL: 0
HOLD: 50
Signal Distribution
00
80
60
40
20
0

Filter Signals

Search Symbol

Filter by Timeframe: All

Filter by Signal: All

Results (50 symbols)

	Symbol	Date	Open	Close	Volume	VolumeChange	Signal_daily	Signal_weekly	Signal_biweekly	Signal_monthly	DCCV_daily
13	SYM0025	2024-12-19 00:00:00	135.108055	134.790033	8326354	-1080621.000000	HOLD	BUY	HOLD	HOLD	0.065400
14	SYM0029	2024-12-19 00:00:00	122.057506	122.710003	4306643	346413.000000	HOLD	HOLD	HOLD	HOLD	0.025500
15	SYM0042	2024-12-19 00:00:00	132.440651	132.223354	2670466	2406392.000000	SELL	HOLD	HOLD	HOLD	-0.879900
16	SYM0001	2024-12-19 00:00:00	211.551932	210.108511	8318391	2127081.000000	HOLD	HOLD	HOLD	HOLD	-0.276100
17	SYM0019	2024-12-19 00:00:00	63.815436	63.937879	609751	-3917138.000000	HOLD	HOLD	HOLD	HOLD	-0.446500
18	SYM0017	2024-12-19 00:00:00	145.925108	147.346722	6721810	521422.000000	HOLD	HOLD	HOLD	HOLD	0.017600
19	SYM0020	2024-12-19 00:00:00	148.611453	149.206175	484890	-2479683.000000	HOLD	HOLD	HOLD	HOLD	-0.344200
20	SYM0034	2024-12-19 00:00:00	94.509092	94.310518	5009579	-4815870.000000	HOLD	HOLD	HOLD	HOLD	0.291300
21	SYM0036	2024-12-19 00:00:00	95.526583	95.793115	8651588	6139983.000000	BUY	HOLD	HOLD	HOLD	0.888200
22	SYM0038	2024-12-19 00:00:00	324.131540	326.955340	1434646	-7593333.000000	HOLD	HOLD	HOLD	HOLD	-0.439600

Symbol	Date	Open	Close	Volume	VolumeChange	Signal_daily	DCCV_daily	Strength_daily	Signal_weekly	DCCV_weekly	Strength_weekly	Signal_biweekly	DCCV_biweekly	Strength_biweekly
SYM0027	12/19/2024	129.73	130.40	8761589	5,73,665	HOLD	0.00	0.03	HOLD	0.28	13.76	HOLD	0.40	19.85
SYM0023	12/19/2024	65.33	65.51	8407610	72,71,934	HOLD	0.48	23.91	HOLD	-0.35	17.72	HOLD	-0.16	7.95
SYM0045	12/19/2024	96.95	97.65	6127547	25,18,550	HOLD	0.46	23.00	HOLD	0.16	8.12	HOLD	0.15	7.45
SYM0000	12/19/2024	92.71	92.95	7938378	11,92,822	HOLD	-0.09	4.72	HOLD	-0.08	4.16	HOLD	-0.38	19.11
SYM0033	12/19/2024	101.18	100.82	526109	-88,56,882	HOLD	0.42	21.15	HOLD	0.31	15.63	HOLD	0.20	10.16
SYM0048	12/19/2024	140.81	140.82	517094	-62,77,064	HOLD	-0.37	18.43	HOLD	-0.17	8.65	HOLD	0.29	14.48
SYM0012	12/19/2024	69.31	69.21	2292977	-12,82,796	HOLD	0.19	9.55	HOLD	-0.04	1.91	HOLD	0.14	7.17
SYM0010	12/19/2024	184.74	186.59	8353359	-9,23,731	HOLD	-0.08	4.12	HOLD	-0.15	7.64	HOLD	-0.34	16.94
SYM0026	12/19/2024	104.59	105.57	5137651	-19,66,434	HOLD	-0.10	5.11	HOLD	-0.03	1.60	HOLD	-0.36	18.19
SYM0035	12/19/2024	98.69	99.08	3810622	15,63,718	HOLD	0.38	18.90	HOLD	0.00	0.08	HOLD	0.12	5.77
SYM0004	12/19/2024	285.00	283.84	8829058	44,31,430	HOLD	-0.40	20.14	HOLD	0.22	10.84	HOLD	0.27	13.70
SYM0018	12/19/2024	158.91	159.53	1578996	-29,70,136	HOLD	-0.35	17.71	HOLD	-0.11	5.72	HOLD	0.16	8.13
SYM0021	12/19/2024	157.29	158.35	2444969	-28,63,227	HOLD	-0.34	16.79	HOLD	-0.04	1.80	HOLD	-0.09	4.65
SYM0025	12/19/2024	135.11	134.79	8326354	-10,80,621	HOLD	0.07	3.27	BUY	0.75	37.34	HOLD	0.12	5.84
SYM0029	12/19/2024	122.06	122.71	4306643	3,46,413	HOLD	0.03	1.28	HOLD	-0.03	1.42	HOLD	-0.37	18.72
SYM0042	12/19/2024	132.44	132.22	2670466	24,06,392	SELL	-0.88	44.00	HOLD	0.11	5.28	HOLD	-0.33	16.29
SYM0001	12/19/2024	211.55	210.11	8318391	21,27,081	HOLD	-0.28	13.80	HOLD	-0.06	3.02	HOLD	-0.07	3.55
SYM0019	12/19/2024	63.82	63.94	609751	-39,17,138	HOLD	-0.45	22.33	HOLD	-0.08	3.89	HOLD	-0.14	6.76
SYM0017	12/19/2024	145.93	147.35	6721810	5,21,422	HOLD	0.02	0.88	HOLD	0.07	3.47	HOLD	-0.15	7.67
SYM0020	12/19/2024	148.61	149.21	484890	-24,79,683	HOLD	-0.34	17.21	HOLD	0.21	10.34	HOLD	0.18	8.94
SYM0034	12/19/2024	94.51	94.31	5009579	-48,15,870	HOLD	0.29	14.56	HOLD	-0.10	5.16	HOLD	0.05	2.71
SYM0036	12/19/2024	95.53	95.79	8651588	61,39,983	BUY	0.89	44.41	HOLD	0.07	3.28	HOLD	0.05	2.65
SYM0038	12/19/2024	324.13	326.96	1434646	-75,93,333	HOLD	-0.44	21.98	HOLD	-0.05	2.27	HOLD	0.15	7.55
SYM0041	12/19/2024	69.38	70.05	3502134	-55,71,345	HOLD	-0.28	14.24	HOLD	-0.11	5.38	HOLD	-0.47	23.56